

PCA on Gene Expression Analysis

Objective

To study gene expression data and identify patterns using Principal Component Analysis (PCA).

Tasks Performed

- Created a scatter plot between XBP1 and GATA3 genes
- Applied PCA and projected the dataset onto the first principal component (PC1)

Dataset Information

- 105 breast cancer samples
- Class labels: ER+ $\rightarrow$ 1, ER- $\rightarrow$ 0

Dataset source: GEO (GSE5325)

Files Used:

filtered.tsv.gz

columns.tsv.gz

class.tsv

Note: Dataset files are not included because of their large size.

Observations

- The scatter plot showed noticeable separation between the two classes.
- PCA successfully captured most of the variation in the first principal component (PC1).

Main Difficulty

Preprocessing the dataset was challenging because the files contained inconsistent formatting, such as additional spaces and extra metadata rows.

Technologies Used

- Python
- Pandas
- Matplotlib
- Scikit-learn